BIFlow BO3 Data Quality Report

Full cleaning run summary generated from on-disk artifacts.

Dataset	iris
Run id	sample_iris_c6d4c3
Generated	2026-10-26 09:00 UTC
Source	iris.csv (c12b0a8f3fee…)
Shape	151 × 5
LLM	LLM off
Config hash	cd2c4238e7f3ab33
Versions	Python 3.14.6, pandas 3.1.0, sklearn 1.9.1, reportlab 5.1.0

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Executive summary

PASS

Rows in	—	Rows out	—
Duplicates removed	—	Cells imputed	0
Outliers flagged	4	Review items	—
Silent deletions	0	Runtime ms	—

0 exact duplicate rows were removed and logged where applicable.
0 cells were imputed using configured deterministic or ML methods.
4 outlier flags were applied on numeric columns per YAML thresholds.
0 items remain in the review queue for human decision.

Safety & integrity checks

Invariant checks derived from parquet files and logs (never hardcoded).

Check	Status	Evidence
Row accounting iris	PASS	in=151 dropped=2 out=149
Dtype preservation iris	PASS	all original dtypes compatible
No literal 'nan' strings iris	PASS	none
Undo round-trip	NOT RUN	No _undo artifacts
Determinism	NOT RUN	No determinism flag in snapshot
Ingest sha256 recorded	PASS	iris.csv c12b0a8f3fee…

Input & parsing

Global parse settings: {}

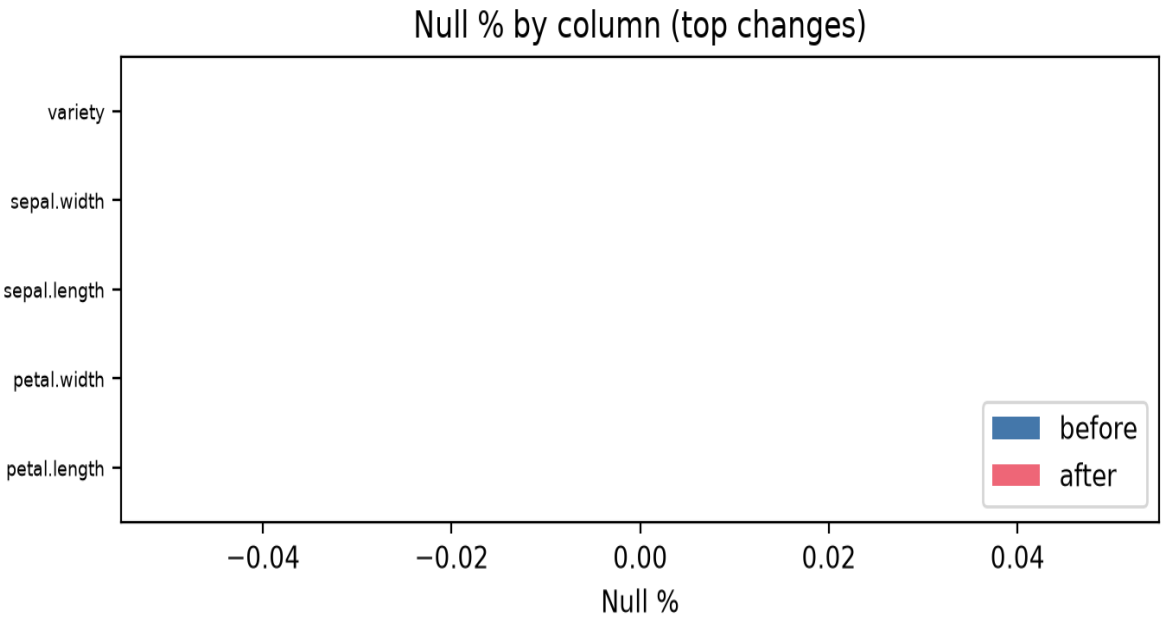
File	Encoding	Delimiter	Rows	Cols	Warnings
iris.csv	auto		151	5	Column 'variety': categorical, 3 values

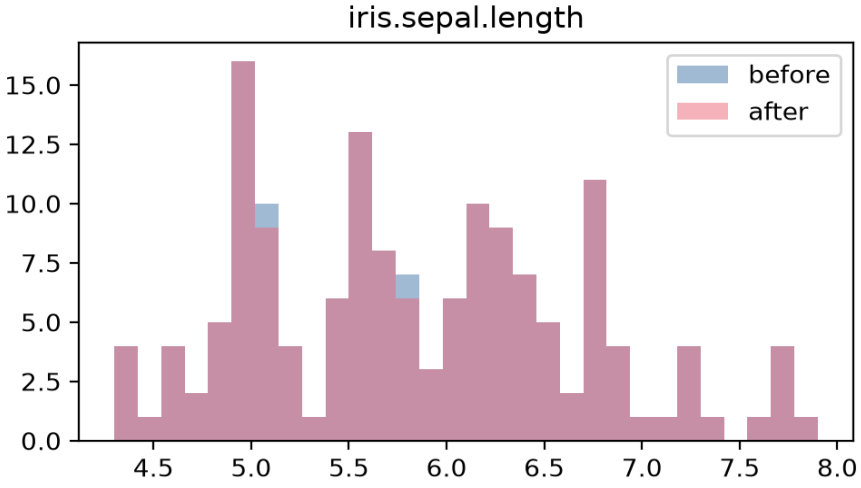
Column recognition

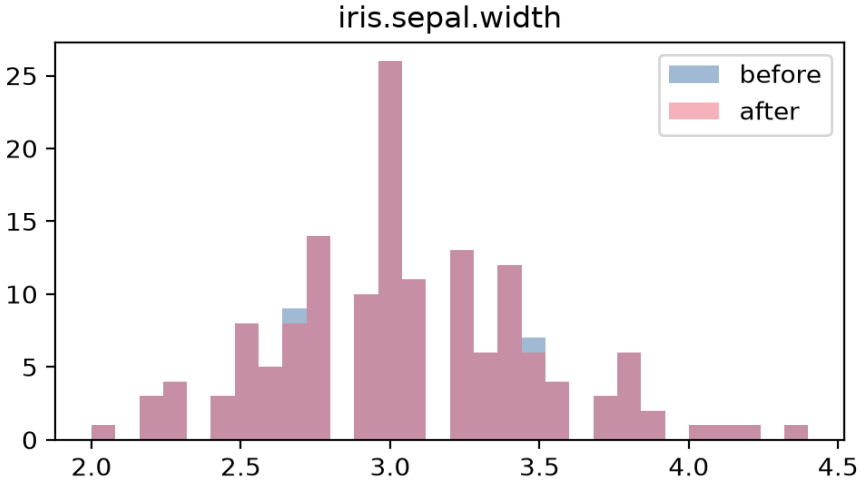
Table	Column	Role	Type	Conf.	Action	Evidence
iris	sepal.length	continuous_numeric	continuous_numeric	0.88	review	continuous numeric role
iris	sepal.width	continuous_numeric	continuous_numeric	0.88	review	continuous numeric role
iris	petal.length	continuous_numeric	continuous_numeric	0.88	review	continuous numeric role
iris	petal.width	continuous_numeric	continuous_numeric	0.88	review	continuous numeric role
iris	variety	categorical	categorical	0.85	review	low/medium cardinality (3 values)

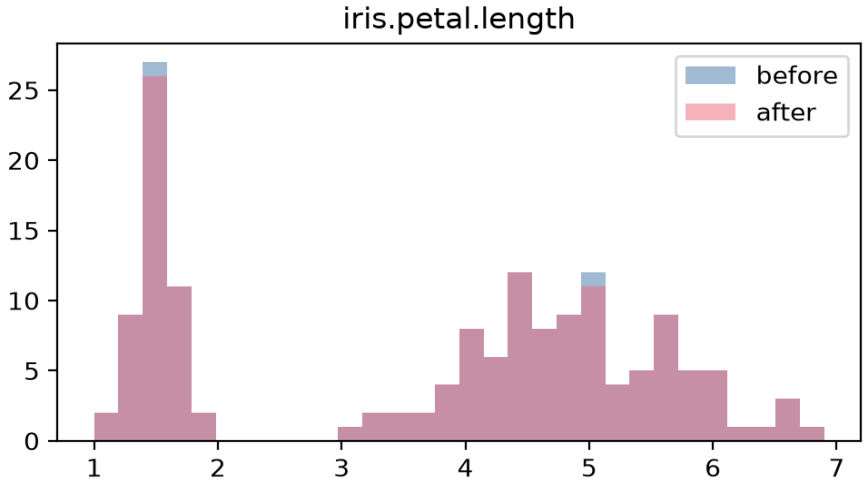
Profile before vs after

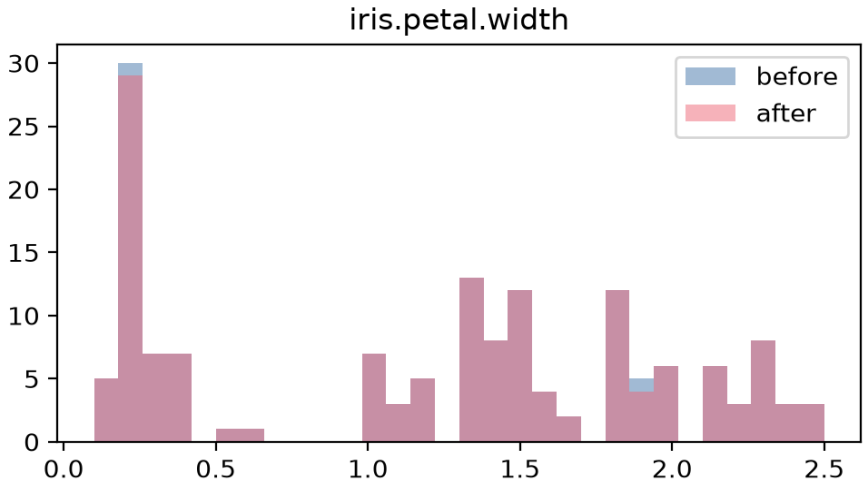
Table	Rows	Dupes	Null cells (sum)
iris	151	2	0


Null % by column (top changes)
variety
sepal.width
sepal.length
petal.width
petal.length
before
after
−0.04
−0.02
0.00
0.02
0.04
Null %


iris.sepal.length
before
after
15.0
12.5
10.0
7.5
5.0
2.5
0.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0


iris.sepal.width
before
after
25
20
15
10
5
0
2.0
2.5
3.0
3.5
4.0
4.5





Cleaning actions

Rule	Method	Columns	Rows affected
detect_outliers	iqr	petal.length, petal.width, sepal.length, sepal.width	4
drop_exact_duplicates	deterministic	None	2

Table	Column	Rule	Rows	Before	After
iris		drop_exact_duplicates	2	—	—
iris	sepal.length	detect_outliers	0	—	—
iris	sepal.width	detect_outliers	4	—	—
iris	petal.length	detect_outliers	0	—	—
iris	petal.width	detect_outliers	0	—	—

Review queue

Nothing needs review

Dropped rows & reconciliation

Table	Rows in	Dropped	Rows out	Status
iris	151	2	149	PASS

Rule	Count	Sample ids
drop_exact_duplicates	2	None, None

Lineage

Step	Method	Rule	Rows
drop_exact_duplicates	deterministic	drop_duplicates(iris)	2
detect_outliers	ml	iqr(iris.sepal.length)	0
detect_outliers	ml	iqr(iris.sepal.width)	4
detect_outliers	ml	iqr(iris.petal.length)	0
detect_outliers	ml	iqr(iris.petal.width)	0



LLM usage

All counts and statistics in this report are read from run artifacts; they never depend on LLM output.

LLM not used

Reproducibility & limitations

Known limitations

- Ambiguous dates go to the review queue; no silent guess.
- No semantic correction of free-text values beyond configured rules.
- Column roles block unsafe numeric imputation on ids and categoricals.
- LLM advisory (when enabled) does not change deterministic counts in this report.
- See README for supported upload formats and issue classes.

Right-to-left scripts (e.g. Arabic) are not shaped; glyphs appear in logical order only.

config_used.yaml (first 120 lines):

```
name: uploaded_iris
domain: upload
source:
type: csv
path: data/uploads/uploaded_iris
tables:
iris: iris.csv
expected_schema:
iris:
sepal.length:
dtype: float
role: continuous_numeric
why: continuous numeric (35 distinct)
sepal.width:
dtype: float
role: continuous_numeric
why: continuous numeric (23 distinct)
petal.length:
dtype: float
role: continuous_numeric
why: continuous numeric (43 distinct)
petal.width:
dtype: float
role: continuous_numeric
why: continuous numeric (22 distinct)
variety:
dtype: str
role: categorical
```

why: categorical, 3 values
relationships: []
cleaning_rules:
primary_keys:
iris: sepal.length
role_inference:
high_null_pct: 60.0
parse_dates: []
trim_strings:
iris:
- variety
drop_exact_duplicates:
- iris
null_strategy:
iris.sepal.length:
strategy: median
allowed:
- median
imputed_flag: sepal.length__imputed
iris.sepal.width:
strategy: median
allowed:
- median
imputed_flag: sepal.width__imputed
iris.petal.length:
strategy: median
allowed:
- median
imputed_flag: petal.length__imputed
iris.petal.width:
strategy: median
allowed:
- median
imputed_flag: petal.width__imputed
iris.variety:
strategy: mode
imputed_flag: variety__imputed
impossible_values: []
outliers:
method: iqr
columns:
- iris.sepal.length
- iris.sepal.width
- iris.petal.length
- iris.petal.width
action: flag
discrete_outliers: false
llm_advisory:
enabled: false
allowed_actions:
- impute_median
- impute_mode
- flag
- ignore

Appendix

Glossary

Term	Meaning
Role	Inferred column purpose (id, categorical, numeric, ...)
Flag	Boolean helper column (__imputed, __outlier, ...)
Imputation	Fill missing values with a documented statistic or model
Outlier	Values flagged by IQR or similar; not always removed
Review queue	Items needing human approval before auto-apply
Lineage	Ordered record of cleaning steps and row counts